

Toward Scalable Quantum Network Management: Unified Formulation, Oracle Design, and Distributed Coordination

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Abstract—This paper presents a unified quantum framework for scalable network management. The framework integrates common optimization modeling, feasibility-preserving oracle design, and distributed coordination. Diverse intra-domain and inter-domain management tasks are represented as assignment problems with different objectives and resource constraints, and the same representation supports their joint management. Unlike penalty-based quantum optimization, the proposed oracle directly encodes the original objective and constraints. This design guarantees feasible accepted solutions without penalty-dependent objective distortion. To address the limited circuit capacity of present-day noisy intermediate-scale quantum processors, large-scale problems are partitioned into local quantum computations with boundary coordination maintaining consistency across local solutions. The framework is implemented for access-point selection and radio-resource allocation and evaluated through noise-free and noisy simulations, quantum hardware execution, and distributed protocol experiments. The results verify exact constraint satisfaction, accurate distribution reconstruction, and scalable coordination, and establish a practical foundation for applying a common quantum computational framework to diverse and coupled network management tasks.

Index Terms—Quantum computing, network management, assignment optimization, oracle design, distributed coordination, hybrid quantum-classical computing

I. INTRODUCTION

Modern wireless networks are evolving from communication infrastructures into programmable systems that integrate communication, computation, and control across diverse services [1]. Network functions now operate on programmable software platforms instead of dedicated hardware and adapt to traffic conditions, wireless environments, and service requirements [2], [3]. As a result, network management involves continuous optimization beyond static configuration. Computational capability has consequently become increasingly important to network operation [2], [4].

Many network management functions share an assignment structure. User association, radio resource allocation, traffic scheduling, task offloading, service placement, and network slicing assign limited resources subject to connectivity, capacity, and service constraints [5], [6]. Their candidate choices, utilities, and resource conditions differ, but each problem seeks a feasible assignment with high network utility. This common structure provides a basis for a computational framework that can address heterogeneous network management tasks and their joint management.

Network scale makes these assignment problems computationally challenging. The number of possible configurations grows combinatorially with network size and the available assignment choices [5]. The computation at a single controller with global information grows with the problem size, while distributed computation introduces the additional need for consistency among local solutions. Scalable network management consequently involves both computation and coordination..

Existing computational approaches address these challenges in different ways. Classical optimization provides rigorous methods for solving constrained network management problems [7], [8]. Approximation algorithms and heuristics trade solution quality or optimality guarantees for reduced computational cost [8], [9]. Artificial intelligence and machine learning provide have greatly improved network automation through traffic prediction, interference management, mobility optimization, and adaptive resource provisioning [2]. However, learned outcomes do not directly solve the constrained assignment problem and generally provide no guarantee of constraint satisfaction [10].

Quantum computing offers a fundamentally different computational approach to these problems. Quantum superposition supports the representation and processing of multiple candidate solutions within a quantum state [10], [17]. Recent advances in quantum hardware and algorithms have expanded their use in communication systems and network management [18], [19]. Quantum techniques, such as quantum search, variational optimization, and QUBO-based formulations have been studied for routing, scheduling, resource allocation, and related combinatorial problems [17], [20]–[22].

Practical network management, however, involves constrained assignment problems in which the network utility defines the objective and operational constraints determine the feasible solutions. Their quantum implementation depends on how these two components are represented in the quantum computation. A common approach incorporates the constraints into the objective through penalty terms. This formulation modifies the original objective, and both constraint satisfaction and the evaluated objective can depend on the selected penalty parameters [20], [21]. Structured oracle methods, including Grover adaptive search (GAS) and quantum tree generator (QTG), provide alternative mechanisms that handle constraints directly during quantum computation.

The representation of the objective introduces another issue. Network utilities are generally computed from physical quantities such as channel conditions, traffic demands, and

resource states and therefore take real values. Existing oracle-based optimization methods represent objective values with finite discrete registers for quantum arithmetic and comparison. Their precision consequently depends on the numerical representation, which can restrict the direct use of real-valued network utilities. A quantum design for network management is requested to account for these utility values without reducing them to a fixed set of discrete coefficients. Quantum hardware further limits the executable scale of network problems. Present-day noisy intermediate-scale quantum (NISQ) processors support limited circuit sizes, whereas a network-wide assignment may involve more variables and constraints than a single quantum circuit can accommodate [18], [23]. Executing such problems with available quantum resources may involve smaller quantum computations and combination of such solutions.

This paper proposes a scalable computational architecture for quantum network management that addresses these issues within a unified computational flow. The original assignment formulation is translated directly into quantum computation. The oracle evaluates the original constraints to identify feasible assignments, while controlled rotations encode real-valued utility coefficients without restricting them to a fixed discrete representation. These operations jointly produce feasible assignments with probabilities determined by their network utilities. For problems beyond the capacity of a single NISQ processor, the computation is partitioned into zone-local quantum problems. Their sampled outputs are coordinated through the UEs shared across zone boundaries to construct a consistent network-wide assignment.

The complete design is implemented in IBM Quantum Computing package and evaluated through ideal and noisy circuit simulations, quantum hardware execution, distributed protocol experiments, and resource-scaling analysis. The evaluation uses wireless network configurations generated by a digital-twin testbed over a real-map environment. The proposed quantum algorithm is executed on a real NISQ processor to verify its operation and exact constraint satisfaction. The results also characterize the reconstruction of the intended utility-dependent distribution, coordination across zone-local quantum computations, and resource scaling. The main contributions of this paper are summarized as follows.

- **Unified quantum framework.** A unified formulation is established to represent heterogeneous network management tasks within a common assignment model and to support their joint formulation. The resulting assignment structure provides a common computational basis for applying quantum computation across these tasks.
- **Formulation-driven quantum design.** A quantum computational design is developed to directly realize the original assignment formulation rather than penalty reformulation. The design guarantees exact constraint satisfaction and directly incorporates real-valued network utilities without conversion to predetermined discrete coefficients. This approach preserves the feasible solution space and the original objective in quantum computation.
- **Scalable implementation and validation.** A distributed quantum architecture is developed to extend network-

wide computation beyond the capacity of individual NISQ processors. Large-scale tasks are organized into coordinated local quantum computations that operate within available quantum resources and collectively construct a consistent network-wide solution. The complete architecture is implemented and validated using real-map wireless network configurations. The proposed quantum computation is also executed on a real quantum processor to verify its operation under actual hardware conditions.

These contributions establish an end-to-end computational framework that connects practical network optimization models to constraint-preserving, utility-aware quantum execution.

The remainder of this paper is organized as follows. Section II reviews related quantum optimization methods for assignment-based network management. Section III presents the unified assignment model and its zone-based decomposition for scalable implementation. Section IV develops the quantum computation for constraint-preserving and utility-weighted assignment and the coordination of distributed quantum outputs. Section V presents the implementation and experimental evaluation, and Section VI concludes the paper.

II. BACKGROUND AND RELATED WORK

This section reviews the common assignment structure of network management problems and quantum approaches for solving constrained assignment problems.

A. Assignment-based network management

Many network management functions can be represented as assignment problems in which communication, radio, and computing resources are assigned under connectivity, capacity, and service constraints [5], [6]. Examples include assigning traffic flows to routes, computing tasks to computing nodes, and service requests to available service locations. Wireless networks exhibit the same assignment structure at different management levels. At the inter-cell level, user equipment (UEs) are assigned to access points (APs) according to link conditions and AP capacities. At the intra-cell level, radio resource blocks (RBs) are assigned to UEs according to resource availability and link utility. Despite differences in assignment choices, utilities, and constraints, these problems share the task of selecting feasible assignments to maximize network utility. This common assignment structure also involves coupled resource constraints, as individual assignment choices can affect multiple conditions. The number of possible assignments grows rapidly with the problem size. These characteristics make constraint handling and scalability important considerations for quantum computation.

B. Quantum approaches to assignment optimization

Quantum approaches provide different mechanisms for representing assignment objectives, enforcing constraints, and searching the resulting solution space. Quantum search algorithms (QSAs) identify desired states through an oracle and amplify their measurement probabilities [12]. Grover's algorithm searches for marked states [13], BBHT algorithm handles an unknown number of marked states [14], and Dürr-Høyer algorithm extends quantum search to minimum or

TABLE I: Comparison of quantum approaches for assignment-based network management. $\bigcirc$: supported; $\triangle$: supported with restrictions; $\times$: not supported.

Approach	Constraint enforcement	Objective preservation	Strict feasibility	Heterogeneous candidates / capacities	Coupled constraints	Reusable quantum design	Distributed execution	Gate-level QPU demo
QUBO, QAOA, quantum annealing [8], [9], [11], [28], [29]	Penalty-augmented objective	$\triangle$	$\times$	$\triangle$	$\triangle$	$\times$	$\triangle$	$\bigcirc$
Quantum learning policies [25]–[27]	Penalty-based training	$\bigcirc$	$\times$	$\triangle$	$\triangle$	$\times$	$\triangle$	$\times$
Variational continuous optimization [30], [31]	Parameterized constraints	$\bigcirc$	$\triangle$	$\times$	$\times$	$\times$	$\times$	$\triangle$
Quantized weighted QSA [19]	Pairwise checks $O(N^2)$	$\times$	$\bigcirc$	$\times$	$\times$	$\times$	$\times$	$\bigcirc$
Enumeration oracle [1]	$O((N+1)^{ \mathcal{U} })$ depth	$\bigcirc$	$\bigcirc$	$\triangle$	$\triangle$	$\times$	$\times$	$\triangle$
Grover adaptive search [21], [22]	One value register per constraint	$\triangle$	$\bigcirc$	$\bigcirc$	$\triangle$	$\triangle$	$\times$	$\bigcirc$
QTG-based knapsack search [23], [24]	$O(N)$ -layer state preparation	$\triangle$	$\bigcirc$	$\bigcirc$	$\times$	$\triangle$	$\times$	$\times$
Proposed	Arithmetic $O(D)$ matches	$\bigcirc$	$\bigcirc$	$\bigcirc$	$\bigcirc$	$\bigcirc$	$\bigcirc$	$\bigcirc$

maximum finding [15]. These methods provide the basis for oracle-driven assignment search [10]. QAOA and quantum annealing represent combinatorial optimization through cost functions [8], [9], [11], whereas quantum machine learning and variational methods use parameterized circuits for learned policies or continuous optimization [25], [26], [30], [31]. Their different representations lead to different ways of handling feasibility, utility, and problem scale.

QAOA and quantum annealing do not directly impose explicit constraints on candidate solutions [8], [9], [11]. Constrained assignment problems are normally converted into QUBO form by adding penalty terms for constraint violations [28], [29]. The resulting cost function combines the original objective with the penalties, making constraint satisfaction dependent on the selected penalty weights. Hybrid decompositions can distribute large QUBO problems across smaller quantum subproblems, but the local problems still use the same penalty-based formulation [28], [29].

Quantum machine learning provides another approach by learning assignment policies through parameterized quantum circuits [25]–[27]. Constraints are incorporated into the training objective or output parameterization instead of being checked directly for each candidate assignment. Strict feasibility is consequently not inherent in the learned policy. Tight constraints can also increase the training effort to suppress infeasible outputs. Network-scale training and execution of parameterized quantum circuits remain difficult on present-day NISQ hardware, and physical quantum processing unit (QPU) demonstrations are generally limited to small circuit instances.

Variational quantum methods have also been applied to continuous network optimization [30], [31]. Their parameterized circuits can optimize continuous control variables without first representing them as discrete assignment states. This property is useful for problems such as beamforming but differs from assignment problems with discrete choices, heterogeneous candidate sets, and multiple capacity constraints. Such methods do not directly provide feasibility checks for individual

candidate assignments under these constraint structures.

Oracle-based QSAs provide a more direct treatment of discrete constraints. Quantized or weighted QSAs can exclude infeasible assignments without adding constraint violations to the objective [19]. Their objective values, however, are represented by a finite set of numerical levels. Different original utilities can be mapped to the same encoded value, which prevents the quantum search from distinguishing the corresponding assignments. Measured assignments with the same encoded value may then need to be compared using their original utilities. Existing constructions are also tailored to specific constraint structures, which limits their use as a common design for heterogeneous assignment problems. Enumeration-based oracles avoid penalty reformulations by checking candidate assignments directly against the original constraints [1]. Accepted states satisfy the specified feasibility conditions. Explicit enumeration of candidate configurations, however, causes circuit depth to grow rapidly with the number of assignment choices and restricts extension to network-scale problems.

More structured oracle methods avoid explicit enumeration. Grover adaptive search (GAS) searches for a feasible assignment with an objective value better than the best value found so far [21], [22]. The current best value serves as a threshold in the oracle, which marks only feasible assignments that improve this threshold. When a better assignment is measured, its objective value becomes the new threshold, and the search is repeated. This iterative progressively identifies better feasible assignments toward the optimum. For heterogeneous assignment problems, the quantum circuit evaluates the objective and multiple resource constraints during this search. Quantum tree generation (QTG) takes a different approach by constructing quantum states that satisfy inequality constraints [23], [24]. For a knapsack problem with a single capacity constraint, the remaining capacity is updated after each assignment choice and represented in the quantum state. This value determines which choices can be added next and prevents assignments that

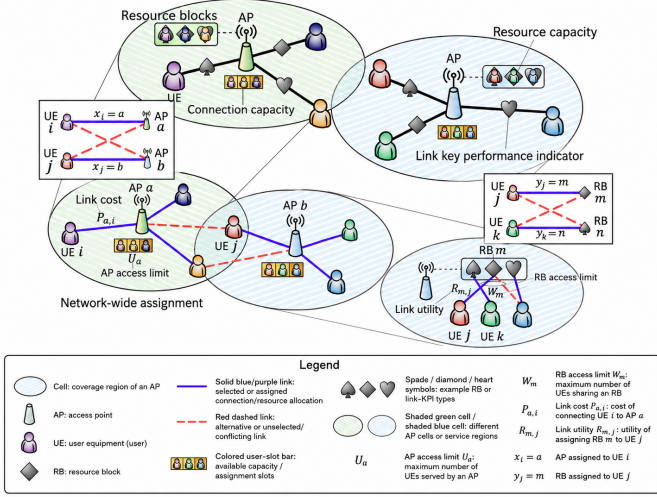


Fig. 1: Assignment structures for wireless network management.

exceed the capacity from being generated. Extending QTG to heterogeneous assignment problems involves multiple capacity constraints and different candidate choices, increasing the state information represented during the generation.

Table I summarizes the reviewed approaches from the perspective of constrained assignment optimization. The comparison focuses on the treatment of constraints and objectives, support for heterogeneous assignment structures, and practical scalability including distributed execution and gate-level QPU implementation. These criteria characterize the computational needs of assignment-based network management rather than general capabilities of each quantum approach. The proposed framework addresses these needs through a unified assignment formulation, formulation-driven quantum computation, and distributed execution. A common quantum design supports exact constraint satisfaction while directly incorporating real-valued network utilities. The computation is further organized into coordinated local quantum problems to support network sizes beyond the capacity of an individual NISQ processor. The next section presents the unified assignment model and its decomposition for scalable execution..

III. UNIFIED ASSIGNMENT MODEL

This section establishes a unified assignment model for inter-cell and intra-cell resource management.

A. Assignment structure and formulation

Fig. 1 illustrates a wireless network configuration with N UEs, M APs, and L RBs. The corresponding sets are denoted by $\mathcal{U}$, $\mathcal{A}$, and $\mathcal{R}$, respectively. The same RBs may be reused across different APs. Under limited coverage of individual APs, UE i can be associated only with APs in its candidate set $\mathcal{A}_i \subseteq \mathcal{A}$. Once associated with AP a , UE i can use any RB in $\mathcal{R}$ at AP i . UE i requests transmission rate d_i , while AP a provides rate capacity W_a . Each RB r can accommodate at most W_r UEs at an AP. The utility of associating UE i with AP a is denoted by $u_{i,a}$, and $w_{i,r}$ denotes the utility for assignment of RB r to UE i . These utilities can represent

link quality, expected rate, load preference, or other service-dependent metric.

Inter-cell management selects a serving AP for each UE under AP coverage and rate-capacity conditions. To represent this association, a binary variable $x_{i,a}$ is defined as

$$x_{i,a} = \begin{cases} 1 & \text{if UE } i \text{ is associated with AP } a, \\ 0 & \text{otherwise.} \end{cases} \quad (1)$$

Each UE is associated with one AP in its candidate set. This requirement is expressed as

$$\sum_{a \in \mathcal{A}_i} x_{i,a} = 1, \quad \forall i \in \mathcal{U}. \quad (2)$$

The total rate demand at AP a is bounded by its capacity as

$$\sum_{i \in \mathcal{U}} d_i x_{i,a} \leq W_a, \quad \forall a \in \mathcal{A}. \quad (3)$$

Since each selected UE contributes its own demand d_i to the AP load, (3) has a knapsack-type capacity structure. The corresponding inter-cell assignment problem is formulated as

$$\begin{aligned} \max_{\mathbf{x}} J_{\text{inter}}(\mathbf{x}) &= \sum_{i \in \mathcal{U}} \sum_{a \in \mathcal{A}_i} u_{i,a} x_{i,a} \\ \text{s.t.} \quad &\sum_{a \in \mathcal{A}_i} x_{i,a} = 1, \quad \forall i \in \mathcal{U}, \\ &\sum_{i \in \mathcal{U}} d_i x_{i,a} \leq W_a, \quad \forall a \in \mathcal{A}, \\ &x_{i,a} \in \{0, 1\}. \end{aligned} \quad (4)$$

Intra-cell management assigns an RB to each UEs served by an AP. Let $\mathcal{U}_a$ denote the UEs associated with AP a . For AP a , the binary RB assignment variable $y_{i,r}$ is defined as

$$y_{i,r} = \begin{cases} 1 & \text{if RB } r \text{ is assigned to UE } i, \\ 0 & \text{otherwise.} \end{cases} \quad (5)$$

Each UE in $\mathcal{U}_a$ is assigned one RB, as described by

$$\sum_{r \in \mathcal{R}} y_{i,r} = 1, \quad \forall i \in \mathcal{U}_a. \quad (6)$$

The occupancy of RB r is bounded by

$$\sum_{i \in \mathcal{U}_a} y_{i,r} \leq W_r, \quad \forall r \in \mathcal{R}. \quad (7)$$

The intra-cell assignment problem at AP a is formulated as

$$\begin{aligned} \max_{\mathbf{y}} J_{\text{intra}}(\mathbf{y}) &= \sum_{i \in \mathcal{U}_a} \sum_{r \in \mathcal{R}} w_{i,r} y_{i,r} \\ \text{s.t.} \quad &\sum_{r \in \mathcal{R}} y_{i,r} = 1, \quad \forall i \in \mathcal{U}_a, \\ &\sum_{i \in \mathcal{U}_a} y_{i,r} \leq W_r, \quad \forall r \in \mathcal{R}, \\ &y_{i,r} \in \{0, 1\}, \quad \forall i \in \mathcal{U}_a, \quad r \in \mathcal{R}. \end{aligned} \quad (8)$$

Although inter-cell association and intra-cell allocation assign different resources, both follow the same assignment structure. Each UE selects one candidate, each selection provides an assignment utility, and the assignments sharing a candidate are subject to its capacity. This common structure

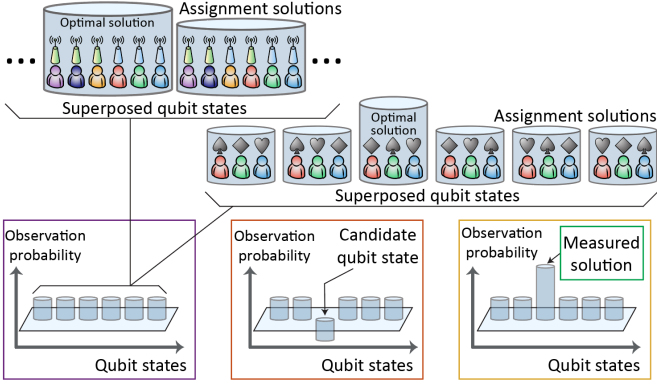


Fig. 2: Quantum representation of candidate assignments with utility-dependent weighting and feasibility screening.

can be represented by a unified assignment model. Let $\mathcal{I}$ denote the items to be assigned and $\mathcal{C}_i$ the candidate set for item i . The set of all candidates is denoted by $\mathcal{C} = \bigcup_{i \in \mathcal{I}} \mathcal{C}_i$. Assignment (i, j) has utility $u_{i,j}$ and consumes $c_{i,j}$ units of the capacity W_j associated with candidate j . For inter-cell association, the candidates correspond to APs and $c_{i,j} = d_i$, whereas intra-cell allocation uses RBs as candidates and $c_{i,j} = 1$. A binary assignment variable is defined as

$$z_{i,j} = \begin{cases} 1 & \text{if item } i \text{ is assigned to candidate } j, \\ 0 & \text{otherwise.} \end{cases} \quad (9)$$

The unified assignment problem is formulated as

$$\begin{aligned} \max_{\mathbf{z}} \quad & J(\mathbf{z}) = \sum_{i \in \mathcal{I}} \sum_{j \in \mathcal{C}_i} u_{i,j} z_{i,j} \\ \text{s.t.} \quad & \sum_{j \in \mathcal{C}_i} z_{i,j} = 1, \quad \forall i \in \mathcal{I}, \\ & \sum_{i: j \in \mathcal{C}_i} c_{i,j} z_{i,j} \leq W_j, \quad \forall j \in \mathcal{C}, \\ & z_{i,j} \in \{0, 1\}, \quad \forall i \in \mathcal{I}, j \in \mathcal{C}_i. \end{aligned} \quad (10)$$

Different assignment-based network management problems can therefore be represented by (10) through their candidate sets, utilities, resource consumption, and capacities.

B. Quantum search strategy for assignment solutions

The unified formulation in (10) defines a search space of candidate assignments, from which feasible assignments with high utility are sought. Quantum computation provides a way to represent these candidates simultaneously in superposition and process their feasibility and utility within the same computational structure. As illustrated in Fig. 2, the assignment constraints distinguish feasible candidate states, and the utility values determine their relative measurement probabilities. The resulting quantum computation searches over feasible assignments while preserving the utility information defined in the original formulation.

As the assignment size increases, representing the entire search space within a single quantum computation becomes difficult under the limited computational resources of present-day NISQ processors. Thus, network-scale execution is ad-

ressed by dividing the unified assignment according to network zones. Each zone contains a subset of APs and the UEs with assignment candidates in that zone. Let $\mathcal{Z}$, $\mathcal{A}_z$, and $\mathcal{U}_z$ denote the set of zones, the APs in zone z , and the UEs included in zone z , respectively. UE i is included in zone z if at least one candidate in $\mathcal{C}_i$ is associated with an AP in $\mathcal{A}_z$. The set of zones containing UE i is denoted by $\mathcal{Z}_i$. A UE with $|\mathcal{Z}_i| > 1$ is defined as a boundary UE, i.e., a UE that belongs to two or more zones. The set of boundary UEs is denoted by $\mathcal{B}$. Each zone contains the assignment variables and capacity constraints associated with its local candidates. For a boundary UE i , the assignment in zone $z \in \mathcal{Z}_i$ is denoted by $z_i^{(z)}$.

The feasible assignments within zone z form the local feasible set $\mathcal{F}_z$. This set contains the assignments that satisfy the assignment and capacity constraints associated with that zone. Local feasibility is represented by

$$\phi_z(\mathbf{z}_z) = \begin{cases} 1, & \text{if } \mathbf{z}_z \in \mathcal{F}_z, \\ 0, & \text{otherwise.} \end{cases} \quad (11)$$

Since a boundary UE is represented in multiple zones, local feasibility alone does not define a valid network-wide assignment. A valid network-wide assignment also maintains consistency for each boundary UE across the zones containing that UE. This consistency is represented by

$$\delta_i(\mathbf{z}_i) = \begin{cases} 1, & \text{if } \mathbf{z}_i^{(z)} \text{ is consistent for all } z \in \mathcal{Z}_i, \\ 0, & \text{otherwise.} \end{cases} \quad (12)$$

The feasibility of a network-wide assignment can then be expressed as

$$\prod_{z \in \mathcal{Z}} \phi_z(\mathbf{z}_z) \prod_{i \in \mathcal{B}} \delta_i(\mathbf{z}_i) = 1. \quad (13)$$

This factorization separates network-wide feasibility into constraint evaluation within individual zones and consistency across their boundaries.

The objective function can also be decomposed according to the same zone partition. For UE i , let $\mathcal{C}_i^{(z)}$ denote the candidates associated with zone z . Each candidate in $\mathcal{C}_i$ belongs to one such local candidate set, and the sets for different zones do not overlap. The utility contribution of zone z is defined as

$$J_z(\mathbf{z}_z) = \sum_{i \in \mathcal{U}_z} \sum_{j \in \mathcal{C}_i^{(z)}} u_{i,j} z_{i,j}. \quad (14)$$

The network-wide objective is therefore recovered by summing the zone contributions as

$$J(\mathbf{z}) = \sum_{z \in \mathcal{Z}} J_z(\mathbf{z}_z). \quad (15)$$

Although a boundary UE appears in multiple zones, each of its assignment candidates belongs to only one zone. Hence, its utility terms are not duplicated in (15). For quantum representation, exponential weighting preserves this additive utility structure across zones. For $\lambda > 0$,

$$\prod_{z \in \mathcal{Z}} \exp[\lambda J_z(\mathbf{z}_z)] = \exp[\lambda J(\mathbf{z})]. \quad (16)$$

Thus, the network-wide utility weight can be constructed from the corresponding zone-level weights while preserving the original objective. The uniqueness of this exponential form under the stated continuity and monotonicity conditions is established in Appendix A. The zone decomposition therefore provides smaller assignment problems for quantum computation while preserving the feasibility and utility structure of the original formulation.

IV. QUANTUM ASSIGNMENT ALGORITHM

This section develops the quantum algorithm that maps the unified assignment formulation to quantum computation. The algorithm directly checks the assignment constraints and represents real-valued utilities through quantum-state amplitudes.

A. Algorithm overview

The proposed algorithm is based on the constraint-screening principle of the constrained quantum framework (CQF) [1]. CQF determines the feasibility of candidate quantum states by directly evaluating the original constraints rather than incorporating constraint violations into the objective. Following this principle, the proposed algorithm evaluates the assignment constraints for each candidate assignment and uses the original real-valued utility to determine its measurement weight. Constraint satisfaction and utility weighting are thereby handled separately without penalty reformulation.

This treatment of feasibility and utility leads to a different computational structure from existing structured oracle methods. GAS searches for an improved solution by marking feasible assignments that exceed a current objective threshold and updating the threshold when a better solution is measured [21]. QTG constructs feasible assignments sequentially, with each candidate choice determined by the remaining resource capacity after the preceding choices [23]. In contrast, the proposed algorithm represents a complete assignment and directly evaluates its feasibility according to the assignment constraints, while its real-valued utility is represented separately through the measurement weight. The same computational structure can be applied to the different assignment problems represented by the unified formulation.

Fig. 3 illustrates the overall quantum assignment procedure. Candidate choices for the UEs are first represented by assignment qubits, such that each quantum state corresponds to a complete candidate assignment. The real-valued utility associated with each assignment is reflected in its quantum amplitude through controlled rotations. The represented assignments are also checked against the candidate-validity and capacity constraints to identify feasible states. Amplitude amplification then concentrates the measurement probability on these feasible states while maintaining their utility-dependent weighting. Repeated executions of the quantum algorithm produce a sample set S_z of feasible assignments for zone z . Each sample contains the assignment choices of the UEs represented in that zone, and the frequency of an assignment reflects its utility-dependent measurement probability. When a zone contains multiple boundary UEs, their assignment choices are recorded within the same sample. The resulting sample set therefore provides both feasible local assignments

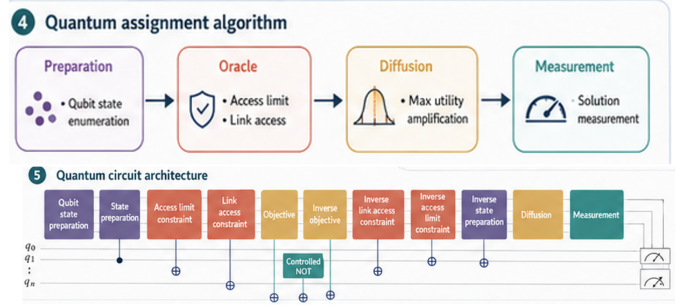


Fig. 3: Quantum assignment structure.

and the boundary assignment information for coordination across zones.

B. Qubit representation of assignment states

The candidate selected by each UE is represented by a binary index. In (10), UE i selects one candidate from C_i . A one-hot encoding of the binary variables $z_{i,j}$ would use one qubit for each candidate. Instead, the proposed representation encodes the index of the selected candidate using $\lceil \log_2 |C_i| \rceil$ qubits. For example, three candidates can be represented by $|00\rangle$, $|01\rangle$, and $|10\rangle$ using two qubits. The remaining basis state $|11\rangle$ has no corresponding candidate and is identified as invalid during feasibility evaluation. The meaning of each valid index is determined by the assignment problem. For inter-cell association, an index identifies an AP in A_i , corresponding to $x_{i,a} = 1$ in (1). For intra-cell allocation, an index identifies an RB, corresponding to $y_{i,r} = 1$ in (5). Thus, different network management problems use different candidate sets within the same index-based representation.

For zone z , let Q_z denote the total number of qubits for representing candidate choices of UEs in U_z . Since UE i is represented by the index of a candidate in $C_i^{(z)}$,

$$Q_z = \sum_{i \in U_z} \lceil \log_2 |C_i^{(z)}| \rceil. \quad (17)$$

A basis state of Q_z qubits specifies one assignment of UEs in U_z . The corresponding real-valued utilities are represented separately through quantum amplitudes.

C. Amplitude representation of assignment utility

The utility of each assignment candidate is represented through its measurement probability, such that a candidate with higher utility receives a larger weight. To construct this representation, one utility qubit is introduced for each UE in addition to the assignment qubits. These utility qubits represent utility-dependent measurement weights and are distinct from the auxiliary qubits used for constraint evaluation. Each utility qubit is initialized to $|0\rangle$ and rotated according to the utility of the selected candidate. An R_Y rotation with angle θ transforms the utility qubit as

$$R_Y(\theta)|0\rangle = \cos \frac{\theta}{2}|0\rangle + \sin \frac{\theta}{2}|1\rangle. \quad (18)$$

The coefficients of $|0\rangle$ and $|1\rangle$ are their probability amplitudes, and the squared amplitudes give the corresponding measurement probabilities. Thus, the probabilities of observing $|0\rangle$

and $|1\rangle$ are $\cos^2(\theta/2)$ and $\sin^2(\theta/2)$, respectively. Selecting the rotation angle according to the candidate utility provides a direct way to map a real-valued utility to a measurement probability. However, the utility values cannot be used directly as measurement probabilities since their numerical range is not restricted to $[0, 1]$. Thus, they are converted into normalized weights while preserving their ordering, such that a candidate with higher utility receives a larger weight. For UE i , the largest utility among its candidates is defined as

$$u_i^{\max} = \max_{j \in \mathcal{C}_i} u_{i,j}. \quad (19)$$

The difference $u_i^{\max} - u_{i,j}$ measures how far candidate j is from the highest-utility candidate. This difference is zero for a highest-utility candidate and becomes larger as the candidate utility decreases. To convert this utility difference into a value between zero and one, the candidate weight is defined as

$$g_{i,j} = \exp[-\lambda(u_i^{\max} - u_{i,j})], \quad (20)$$

where $\lambda > 0$ controls how strongly utility differences are reflected in the weights. This mapping gives $g_{i,j} = 1$ for a highest-utility candidate and $0 < g_{i,j} < 1$ for candidates with lower utility. Furthermore, $g_{i,j}$ increases monotonically with $u_{i,j}$, and the ordering of the original utility values is preserved. The weight $g_{i,j}$ can now be represented by the measurement probability of the utility qubit. From (18), the probability of observing $|0\rangle$ after the rotation is $\cos^2(\theta/2)$. The rotation angle for candidate j is selected to satisfy

$$\theta_{i,j} = 2 \arccos \sqrt{g_{i,j}}. \quad (21)$$

When candidate j is selected for UE i , this rotation transforms the utility qubit as

$$R_Y(\theta_{i,j})|0\rangle = \sqrt{g_{i,j}}|0\rangle + \sqrt{1 - g_{i,j}}|1\rangle. \quad (22)$$

The utility qubit is observed in $|0\rangle$ with probability $g_{i,j}$. A candidate with higher utility has a larger $g_{i,j}$ and consequently contributes a larger measurement weight.

The same construction is applied to the candidate selected by each UE. For a complete assignment $\mathbf{z}$, consider the event in which all utility qubits are measured in $|0\rangle$. Since UE i selects exactly one candidate, the probability associated with this event contains one weight $g_{i,j}$ from each UE, given by

$$G(\mathbf{z}) = \prod_{i \in \mathcal{I}} \prod_{j \in \mathcal{C}_i} g_{i,j}^{z_{i,j}}. \quad (23)$$

Substituting (20) into (23) gives

$$G(\mathbf{z}) = \exp \left[\lambda \sum_{i \in \mathcal{I}} \left(\sum_{j \in \mathcal{C}_i} u_{i,j} z_{i,j} - u_i^{\max} \right) \right] = \kappa \exp[\lambda J(\mathbf{z})], \quad (24)$$

where

$$\kappa = \exp \left[-\lambda \sum_{i \in \mathcal{I}} u_i^{\max} \right] \quad (25)$$

is independent of the selected assignment. Note that (24) establishes the relation between the original assignment objective and its quantum representation. Since κ is common to all assignments, an assignment with a larger $J(\mathbf{z})$ has a larger

measurement weight. Therefore, the exponential mapping preserves the ordering of the original objective while converting real-valued utility coefficients into probabilities that can be represented through qubit rotations.

For inter-cell association, candidate j corresponds to an AP $a \in \mathcal{A}_i$, and $u_{i,j}$ becomes the association utility $u_{i,a}$. For intra-cell allocation, candidate j corresponds to an RB $r \in \mathcal{R}$, and $u_{i,j}$ becomes the allocation utility $w_{i,r}$. In both cases, the selected candidate determines the controlled rotation through its utility coefficient. This candidate-wise mapping provides a common utility representation for both problems.

The same utility representation is applied independently to the candidates contained in each zone. For zone z , the controlled rotations correspond to the utility terms included in $J_z(\mathbf{z}_z)$. The resulting assignment weight is proportional to $\exp[\lambda J_z(\mathbf{z}_z)]$. Since the network-wide objective is the sum of the zone contributions, (16) shows that the product of these zone-level weights recovers the weight associated with the original objective. The utility representation is preserved when the assignment problem is divided across zones.

To prepare the assignment states in zone z , a Hadamard gate is applied to each of the Q_z assignment qubits, placing their candidate indices in superposition. The assignment qubits then control the utility rotations corresponding to the represented candidates. Let Θ_z denote the combined utility-rotation operation in zone z . The resulting state-preparation operation is

$$\mathcal{P}_z = \Theta_z H^{\otimes Q_z}. \quad (26)$$

The prepared quantum state represents the candidate assignments together with their utility-dependent weights. The subsequent operation evaluates these states according to the assignment constraints.

D. Constraint evaluation and feasibility marking

The states prepared by (26) represent candidate assignments with utility-dependent measurement probabilities. Feasibility is determined from two conditions: each binary index represents an admissible candidate, and the resulting assignment satisfies capacity constraints. These conditions are evaluated without measuring the assignment qubits, preserving the superposition of the represented assignments.

Fig. 4 shows the validity evaluation of the binary assignment representation. According to (17), the choice of UE i is represented using $\lceil \log_2 |\mathcal{C}_i| \rceil$ qubits. For example, if UE i can select one of three RBs, two qubits provide four binary states. Three states, $|00\rangle$, $|01\rangle$, and $|10\rangle$, represent available RB choices, while the remaining state $|11\rangle$ does not correspond to an RB and is therefore invalid. Each UE i is associated with a validity flag qubit, and its state after the evaluation is denoted by $|v_i\rangle$. The flag is initialized to $|0\rangle$ and changed to $|1\rangle$ if the binary index of UE i is valid as

$$|v_i\rangle = \begin{cases} |1\rangle & \text{if the index is valid,} \\ |0\rangle & \text{otherwise.} \end{cases} \quad (27)$$

This evaluation leaves the assignment qubits unchanged.

Capacity constraints are evaluated by accumulating the resource usage specified by an assignment and comparing the accumulated value with the corresponding capacity. An

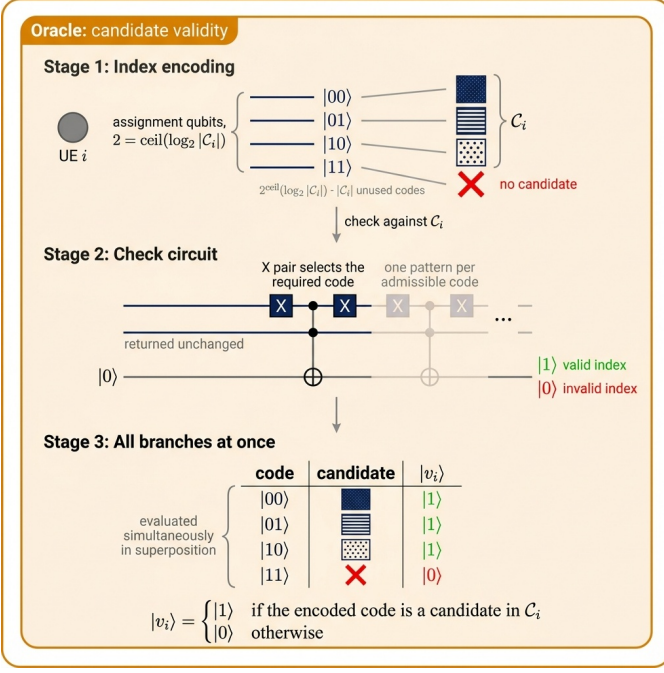


Fig. 4: Validity evaluation of the binary assignment representation.

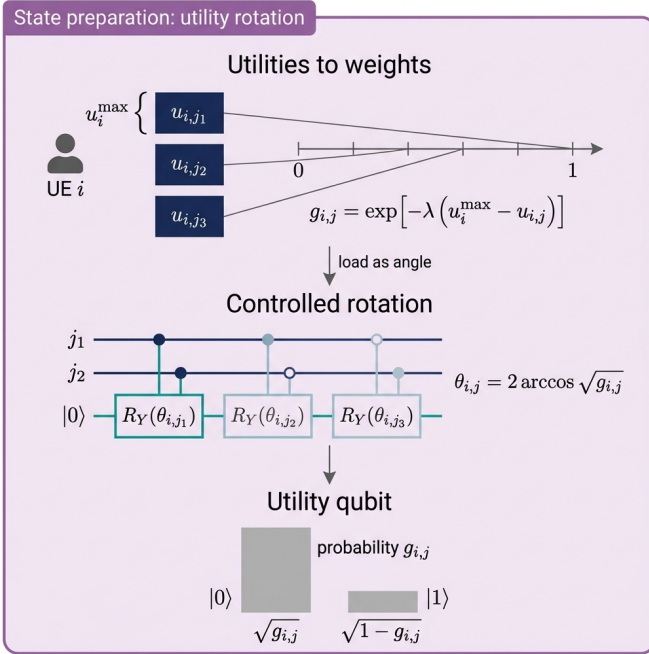


Fig. 5: Quantum evaluation of an AP rate-capacity constraint.

auxiliary register initialized to $|0\rangle$ is used for the accumulation, and the comparison result is recorded in a capacity flag qubit.

Fig. 5 shows this evaluation for the AP rate-capacity constraint in inter-cell association. For AP a , the assignment qubits control the accumulation of the requested rates d_i in the auxiliary register. The accumulation is implemented using a quantum Fourier transform (QFT)-based adder. The QFT maps the auxiliary register to the Fourier basis, where the addition is computed through controlled phase rotations without explicit

carry propagation. When a value x is added to a Q_z^{cnt} -qubit auxiliary register in zone z , the b -th qubit undergoes the phase rotation as

$$\varphi_b(x) = 2\pi \frac{x}{2^b}, \quad b = 1, \dots, Q_z^{\text{cnt}}, \quad (28)$$

where $\varphi_b(x)$ denotes the corresponding rotation angle. For AP a , x corresponds to the requested rate d_i , and the rotation $\varphi_b(d_i)$ is applied only when UE i selects AP a . The rotations contributed by the assigned UEs accumulate as

$$\sum_{i \in \mathcal{U}} x_{i,a} \varphi_b(d_i) = 2\pi \frac{\sum_{i \in \mathcal{U}} d_i x_{i,a}}{2^b} = \varphi_b\left(\sum_{i \in \mathcal{U}} d_i x_{i,a}\right). \quad (29)$$

The accumulated phases represent the total requested rate assigned to AP a in the Fourier basis. In this representation, the accumulated rate is encoded in the phases of the auxiliary qubits rather than as a binary value. Since the capacity comparison uses the binary value of the accumulated rate, the inverse QFT converts the phase representation back to the computational basis. The resulting auxiliary-register state is

$$|\text{aux}_a\rangle = \left| \sum_{i \in \mathcal{U}} d_i x_{i,a} \right\rangle, \quad (30)$$

where $|\text{aux}_a\rangle$ denotes the state of the auxiliary register representing the accumulated rate. A quantum comparator compares this value with the AP capacity W_a . The resulting capacity flag $|f_a\rangle$ is given by

$$|f_a\rangle = \begin{cases} |1\rangle & \text{if } \sum_{i \in \mathcal{U}} d_i x_{i,a} \leq W_a, \\ |0\rangle & \text{otherwise.} \end{cases} \quad (31)$$

The evaluation for the RB occupancy constraint of intra-cell allocation is shown in Fig. 6. The same QFT-based accumulation is applied with a unit value for each UE assigned to RB r , resulting in the comparator input expressed as

$$|\text{aux}_r\rangle = \left| \sum_{i \in \mathcal{U}_a} y_{i,r} \right\rangle, \quad (32)$$

which represents the number of UEs assigned to RB r . The comparator compares this value with the occupancy limit W_r . The resulting capacity flag $|f_r\rangle$ is given by

$$|f_r\rangle = \begin{cases} |1\rangle & \text{if } \sum_{i \in \mathcal{U}_a} y_{i,r} \leq W_r, \\ |0\rangle & \text{otherwise.} \end{cases} \quad (33)$$

For both capacity evaluations, the auxiliary register needs sufficient width to represent the accumulated value before comparison. A Q_z^{cnt} -qubit register represents values modulo $2^{Q_z^{\text{cnt}}}$, and an insufficient register width can cause the accumulated value to wrap around. To prevent this, the register width is selected such that

$$2^{Q_z^{\text{cnt}}} > \max \left\{ \max_{a \in \mathcal{A}_z} \sum_{i \in \mathcal{U}_z} d_i, |\mathcal{U}_z| \right\}, \quad (34)$$

where the first term bounds the accumulated rate for an AP, while the second bounds the number of UEs that can be assigned to an RB. The auxiliary register can represent the values used in both capacity comparisons without wraparound. After each capacity evaluation, the accumulation operations

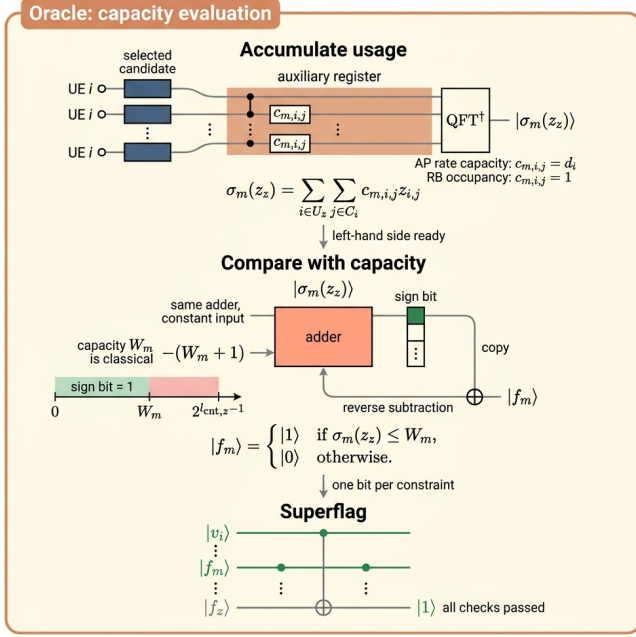


Fig. 6: Quantum evaluation of an RB occupancy constraint.

are reversed to restore the auxiliary register to $|0\rangle$. This uncomputation makes the register available for another capacity constraint while preserving the comparison result in the corresponding capacity flag. Thus, the same auxiliary register can be reused to evaluate multiple capacity constraints.

The validity flags $|v_i\rangle$ and the capacity flags collectively determine whether a represented assignment is feasible in zone z . The results of these evaluations are combined into a single feasibility qubit, referred to as the superflag. The superflag is set to $|1\rangle$ only when $|v_i\rangle = |1\rangle$ for every $i \in \mathcal{U}_z$ and all applicable capacity flags are also in $|1\rangle$. Its state is given by

$$|f_z\rangle = \begin{cases} |1\rangle & \text{if all conditions are satisfied,} \\ |0\rangle & \text{otherwise.} \end{cases} \quad (35)$$

This single flag provides a unique control for marking feasible assignments. In contrast, applying a phase change independently for each constraint can fail to distinguish feasibility when multiple constraints are violated, since an even number of phase changes cancels. Proposition 1 states this property.

Proposition 1 (Strict single-superflag marking). *A phase mark controlled by the superflag distinguishes a feasible assignment from every infeasible assignment, independent of the number of violated conditions.*

Proof: If each of V violated conditions independently applies a phase change, the resulting phase factor is $(-1)^V$. This factor equals one when $V = 0$, but also when V is any positive even integer. Individual phase marking cannot uniquely distinguish feasibility. The superflag instead combines all constraint results before phase marking and reaches its marking state only when every condition is satisfied. ■

The superflag is combined with utility qubits to mark feasible assignment branches according to the utility representation.

For each UE i , let $|0\rangle_i$ denote the $|0\rangle$ state of its utility qubit. As in (22), the probability associated with this state depends on the utility of the selected candidate. The phase-marking operation Λ_z changes the phase only when the superflag is $|1\rangle$ and all utility qubits are in $|0\rangle$. Its action is defined by

$$\Lambda_z \left(|1\rangle_{f_z} \bigotimes_{i \in \mathcal{U}_z} |0\rangle_i \right) = -|1\rangle_{f_z} \bigotimes_{i \in \mathcal{U}_z} |0\rangle_i. \quad (36)$$

The superflag restricts this phase marking to feasible assignments, while the utility-qubit amplitudes determine their relative weights. According to (24), a feasible assignment with a larger objective value has a larger weight in the marked branch.

The zone oracle combines the constraint evaluation with the phase marking. Let $\mathcal{E}_z$ denote the validity and capacity evaluation for zone z . After the phase marking Λ_z , $\mathcal{E}_z^\dagger$ reverses this evaluation to restore the auxiliary registers and flags while preserving the phase mark. The zone oracle is expressed as

$$\mathcal{O}_z = \mathcal{E}_z^\dagger \Lambda_z \mathcal{E}_z. \quad (37)$$

The resulting oracle marks only the assignment branches that satisfy the original assignment constraints, while their relative measurement probabilities remain determined by the utility-dependent amplitudes. Feasibility and utility are represented within the same quantum computation without converting constraint violations into penalty terms.

E. Amplification and measurement of feasible states

The constraint evaluation identifies feasible assignments, and the oracle marks their corresponding quantum-state components by a phase change. This marking distinguishes feasible assignments from the remaining states without changing their measurement probabilities. Amplitude amplification combines the oracle with a reflection operation to increase the probability of observing the marked assignments. The procedure is applied independently to each zone. The probability associated with the marking condition also depends on the utility of each assignment. As defined in (22), the utility qubit of UE i has probability $g_{i,j}$ of being measured in $|0\rangle$ when candidate j is selected. For assignment z in zone z , the probability that the utility qubits of all UEs are measured in $|0\rangle$ is given by

$$g_z(z) = \prod_{i \in \mathcal{U}_z} \prod_{j \in \mathcal{C}_i^{(z)}} g_{i,j}^{z_{i,j}}. \quad (38)$$

For inter-cell association, candidate j corresponds to an AP $a \in \mathcal{A}_i$. The weight $g_{i,j}$ is obtained from the association utility $u_{i,a}$ of UE i and AP a . For intra-cell allocation, candidate j corresponds to an RB $r \in \mathcal{R}$, with $g_{i,j}$ obtained from the allocation utility $w_{i,r}$ of UE i and RB r .

The Q_z assignment qubits in (17) assign an initial probability of 2^{-Q_z} to each represented binary assignment. The oracle applies the phase mark when the assignment belongs to $\mathcal{F}_z$ and all corresponding utility qubits are in $|0\rangle$. The total probability associated with this marking condition before amplification is

$$\mu_z = \frac{1}{2^{Q_z}} \sum_{z \in \mathcal{F}_z} g_z(z). \quad (39)$$

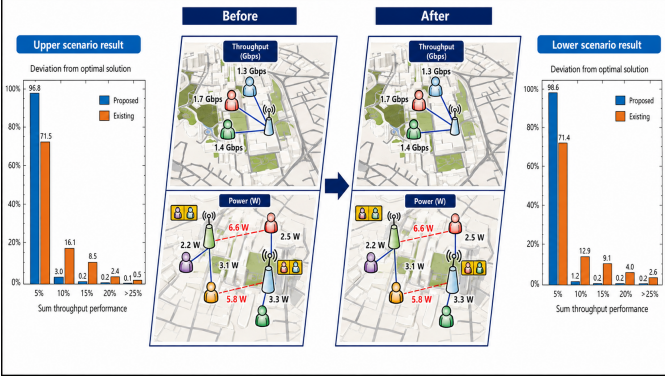


Fig. 7: Quantum implementation of one amplitude-amplification round.

The probability of a particular feasible assignment $\mathbf{z} \in \mathcal{F}_z$, conditioned on this marking condition, is

$$p_z(\mathbf{z}) = \frac{2^{-Q_z} g_z(\mathbf{z})}{\mu_z} = \frac{\exp[\lambda J_z(\mathbf{z})]}{\sum_{\mathbf{z}' \in \mathcal{F}_z} \exp[\lambda J_z(\mathbf{z}')]} \quad (40)$$

A feasible assignment with a larger local utility $J_z(\mathbf{z})$ has a larger probability within this conditional distribution.

Fig. 7 shows one round of amplitude amplification. The zone oracle $\mathcal{O}_z$ applies the phase mark according to the marking condition, followed by a reflection about the state prepared by $\mathcal{P}_z$ in (26). Let $\mathcal{D}_z$ denote this reflection. One amplification round is expressed as

$$\begin{aligned} \mathcal{G}_z &= \mathcal{D}_z \mathcal{O}_z, \\ \mathcal{D}_z &= \mathcal{P}_z (2|0\rangle\langle 0| - I) \mathcal{P}_z^\dagger, \end{aligned} \quad (41)$$

where I denotes the identity operator. The reflection is constructed with respect to the state prepared by $\mathcal{P}_z$. The operation $\mathcal{P}_z^\dagger$ first maps this prepared state back to the reference state $|0\rangle$. A reflection about this reference state is then implemented by $2|0\rangle\langle 0| - I$. This operation leaves the component along $|0\rangle$ unchanged with a phase factor of $+1$, while reversing the sign of every orthogonal component with a phase factor of -1 . The relative phase reversal reflects the state about the $|0\rangle$ direction. Applying $\mathcal{P}_z$ maps this reflection back to the original state space, such that $\mathcal{D}_z$ becomes a reflection about the state prepared by $\mathcal{P}_z$. When combined with the phase change introduced by $\mathcal{O}_z$, this second reflection changes the amplitudes in a direction that increases the contribution of the states satisfying the marking condition.

The amplification round $\mathcal{G}_z$ in (41) can be repeated to increase this contribution. With k amplification rounds following the state preparation in (26), the complete quantum operation for zone z is expressed as

$$\mathcal{C}_z = \mathcal{G}_z^k \mathcal{P}_z. \quad (42)$$

Let $P_z(k)$ denote the total probability associated with the marking condition after k amplification rounds. For the initial marking probability μ_z in (39), this probability is given by

$$P_z(k) = \sin^2[(2k+1) \arcsin \sqrt{\mu_z}]. \quad (43)$$

Amplitude amplification changes the total probability associ-

ated with the marking condition from μ_z to $P_z(k)$ without changing $p_z(\mathbf{z})$ in (40). A feasible assignment with a larger $J_z(\mathbf{z})$ remains more likely to be measured than one with a smaller $J_z(\mathbf{z})$. After amplification, assignment and utility qubits are measured. A measurement is accepted when all utility qubits are observed in $|0\rangle$ and the decoded assignment belongs to $\mathcal{F}_z$. Repeated circuit executions produce multiple feasible local assignments according to $p_z(\mathbf{z})$. Assignments with larger local utilities appear more frequently in these measurements, providing utility-ranked feasible solutions for subsequent coordination across zones.

F. Coordination of local quantum outputs

For coordination across zones, the feasible assignments obtained from the repeated measurements are collected into the joint sample set $\mathcal{S}_z$ for each zone z . Each sample specifies the assignments of all UEs represented in that zone, preserving the relationships among their candidate choices. These joint samples provide the boundary-assignment information used for coordination.

Let $K_z = |\mathcal{S}_z|$, and let $\mathbf{z}_z^{(n)}$ denote the n -th accepted sample in zone z . For boundary UE i represented in zone $z \in \mathcal{Z}_i$, the empirical probability of candidate j is

$$\pi_{z,i}(j) = \frac{\sum_{n=1}^{K_z} w_z^{(n)} z_{i,j}^{(n)}}{\sum_{n=1}^{K_z} w_z^{(n)}}, \quad j \in \mathcal{C}_i, \quad (44)$$

where $w_z^{(n)}$ adjusts the sample weight when the local sampling exponent differs from the common coordination exponent. When sampling exponents are identical, all sample weights are equal, and (44) reduces to the relative frequency of candidate j in $\mathcal{S}_z$. The reweighting procedure is detailed in Appendix B.

For each boundary UE, candidate probabilities from zones containing that UE are combined into the consensus distribution as

$$b_i(j) = \frac{\prod_{z \in \mathcal{Z}_i} \pi_{z,i}(j)}{\sum_{j' \in \mathcal{C}_i} \prod_{z \in \mathcal{Z}_i} \pi_{z,i}(j')}. \quad (45)$$

A candidate receives a larger consensus probability when it is consistently supported across zones containing UE i . The reduction of zone-local distributions to their joint boundary distributions is established in Appendix B.

The boundary assignments are resolved sequentially through confidence-ordered decimation. Among unresolved boundary UEs, UE with the highest confidence in its consensus distribution is selected and assigned to the candidate with the largest consensus probability. Each affected zone then filters $\mathcal{S}_z$ to include only the joint samples consistent with this assignment. Since each sample jointly represents the assignments of UEs in that zone, filtering one boundary UE can change candidate probabilities of unresolved boundary UEs. Their marginal and consensus probabilities are recomputed from filtered sample sets before another boundary UE is selected. Filtering the joint samples preserves the assignment relationships represented within each local sample while progressively imposing resolved boundary assignments. If the effective sample size of an affected zone becomes insufficient, only that zone is sampled again with resolved boundary assignments fixed. The

confidence measure, filtering rule, and selective resampling procedure are detailed in Appendix B-C. The procedure terminates when all boundary assignments are consistent across their corresponding zones. The remaining local assignments satisfy their zone-level feasibility conditions, while resolved boundary assignments satisfy the consistency condition in (13). The coordinated local assignments therefore form a feasible network-wide assignment.

V. NUMERICAL EVALUATION

This section verifies the zone-local distribution, measures it under noise, evaluates boundary coordination on a partitioned region, and characterizes scaling with network size. Implementation, generator, seeds, cached data, noise models, and plotting scripts are in the project repository.¹

A. Setup

The evaluation instantiates (10) for joint AP–RB assignment. Each candidate $j = (a, r) \in \mathcal{C}_i$ loads two resources at once: the admission budget of AP a , with $c_{m,i,j} = d_i$ and $W_m = W_a$, and the occupancy of RB r , with $c_{m,i,j} = 1$ and $W_m = W_r$. The oracle of Section IV-D evaluates both on the same assignment.

APs sit on a jittered unit grid of side g , measured in AP spacings, with two UEs and four RBs per AP, a coverage radius of 1.2 spacings, and at most two joint candidates per UE, so $|\mathcal{C}_i| \leq 2$ and $\ell_i = 1$. The limits are $W_r = 1$, $W_a = 4$, and $d_i = 1$, and utilities decay with UE–AP distance with a per-RB perturbation. Only g varies, holding density, candidate degree, and capacity profile fixed. Greedy region growing with single-AP boundary refinement partitions the AP co-coverage graph, minimizing $|\mathcal{B}|$ subject to the partition budget $G_z \leq 2000$ for one oracle $\mathcal{O}_z$.

The target common exponent is $\beta = 1.5$, so $\lambda = \beta/\bar{u}$ with $\bar{u}$ the mean candidate utility. Each zone applies $k = 1$ round in (42) and reports $K_z = 200$ accepted joint samples, with re-sampling triggered when $\tilde{K}_z < K_{\min} = 25$. A report costs about $K_z/P_z(k)$ executions, with $P_z(k) \simeq (2k+1)^2\mu_z$ here, so one round gives roughly a ninefold reduction. The round count maximizing $P_z(k)$ ranges from about five for the sparsest count to about 900 for the most tightly constrained, motivating the practical choice $k = 1$.

To respect a per-zone budget of 10^4 executions, zone z runs at the largest exponent $\beta_z \leq \beta$ that meets the budget at $k = 1$, and its samples are reconstructed at the common exponent using the reweighting in Appendix B-B. Across zones, β_z ranges from 0.31 to 1.50 and $\tilde{K}_z$ stays at or above 118 of 200 retained samples.

B. Zone-local distributions, exactly and under noise

Six zones of at most seven UEs and 24 qubits admit statevector checks. The largest total-variation distance from $p_z(z) \propto \exp[\lambda J_z(z)]$ on $\mathcal{F}_z$ is 2.7×10^{-15} with no accepted probability outside $\mathcal{F}_z$, so the rotations realize the weight (16) requires and the superflg admits exactly the feasible set.

Under a device noise model, transpiled to its basis gates and coupling map with 10^4 shots per zone, the accepted-branch mass falls to 52% of its noise-free value and the distance from p_z is 0.094, against 0.061 noise-free at the same $K_z = 200$; both trace to the transpiled depth of the QFT accumulators and the superflg conjunction.

Strict feasibility survives unchanged. The acceptance test of Section IV-E applies to the decoded assignment rather than the internal flags, so corrupted outcomes are discarded: the discard rate rises from 12% to 29%, and no infeasible assignment entered any zone report.

Since (26) is a product over UEs before screening and k enters only through $P_z(k)$, independent per-UE utility-weighted sampling followed by feasibility screening induces the same accepted-branch law; the experiments below use this exact sampler and account execution cost separately.

C. One partitioned region

Fig. 8 follows one campus instance of $M = 25$ APs and $N = 50$ UEs. It is drawn from a family of 16 seeds on the same map, 14 of which are globally feasible, and the statistics below are over those 14. Throughout, J^* is the centralized strict optimum from HiGHS. The partition gives $|\mathcal{Z}| = 14$ zones within budget and $|\mathcal{B}| = 23$ boundary UEs, so 46% of the UEs require coordination, and the four displayed zone distributions match their exact accepted laws to 0.061 with no mass outside $\mathcal{F}_z$ despite differing β_z .

On the two most dependent boundary-UE pairs the product of the marginals places 19% and 23% of its belief on joint choices that violate a capacity condition in $\mathcal{M}_z$ and that no sample in $\mathcal{S}_z$ realizes. Against a one-pass baseline identical in sampling and commitment rule, re-conditioning wins on 13 of the 14 instances by 1.2 percentage points of J^* . Five re-samples add 18% to the region’s execution count. The result satisfies every capacity and consistency condition and reaches 99.2% of J^* , the family mean, inside the observed 98.1–100% range.

D. What one zone circuit costs

For zone z , let $D_z = \sum_{i \in \mathcal{U}_z} |\mathcal{C}_i|$ denote the number of evaluated candidate edges, $\ell_{\max,z} = \max_{i \in \mathcal{U}_z} \lceil \log_2 |\mathcal{C}_i| \rceil$ the widest candidate code, and $\ell_{\text{cnt},z}$ the load-counter width. With approximate-QFT arithmetic and standard multi-control decompositions, the modeled depth T_z and two-qubit count G_z of one oracle $\mathcal{O}_z$ scale as

$$T_z, G_z = O(D_z (\ell_{\max,z} + \ell_{\text{cnt},z}) + |\mathcal{M}_z| \ell_{\text{cnt},z}). \quad (46)$$

Under bounded candidate degree and fixed counter width, both terms grow linearly with the zone size. Extra counters parallelize only the accumulation term, so adding counters has a floor.

Fig. 9 sets $D_z = 2N_z$, $\ell_{\text{cnt},z} = 3$, and $|\mathcal{M}_z| = N_z$ for all quantum curves. Depths convert to logical latency as $\tau_z = T_z t_{\text{layer}}$ with $t_{\text{layer}} = 12.5$ ns, and the enhanced schedule uses $R_{\text{cnt}} = 4$ parallel counters. Each method is charged one local pass, excluding repeated search rounds, accepted-draw repetitions, readout, queuing, and boundary coordination.

¹https://github.com/gyh1238/SCQF/tree/main/quantum_transaction

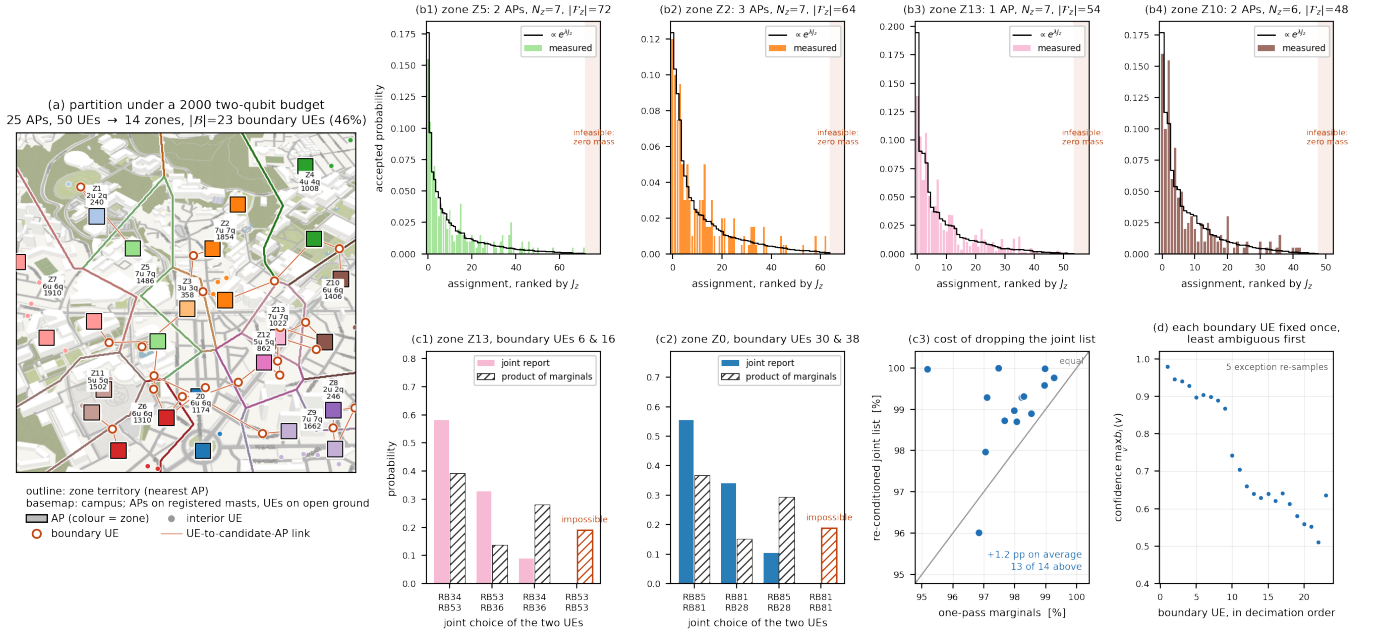


Fig. 8: Zone-local sampling and boundary coordination.

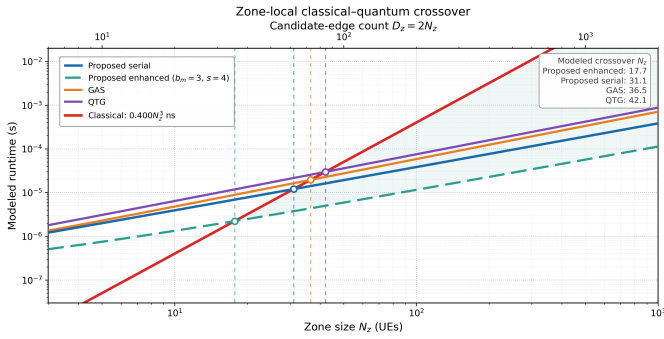


Fig. 9: Zone-local computational cost.

The classical reference is ρN_z^3 ns with $\rho = 0.400$, inside the 0.058–0.740 range measured for the Hungarian method in [1]. It favors the classical side because that method omits the capacity constraints evaluated here. Crossovers occur at $N_z = 17.7$ for the enhanced schedule, 31.1 for the serial schedule, 36.5 for GAS, and 42.1 for QTG. The current partition produces zones of 1–13 UEs; extending the largest measured circuit to $N_z \approx 18$ requires about 2700 gates, or 35% above the current 2000-gate budget.

E. What happens as the region grows

The region side is varied over $g = 3, \dots, 9$ with six seeds per size; two of the 42 instances are globally infeasible and dropped, and the remaining 40 span $N = 18$ to $N = 162$ UEs and $|Z| = 5.3$ to 45.0 zones on average, so the network grows ninefold at fixed local density. A centralized circuit for the whole region grows from 10801 to 173983 two-qubit gates, a factor of 16, while the largest zone circuit grows from 1693 to 1950, a factor of 1.15, never leaving the partition budget:

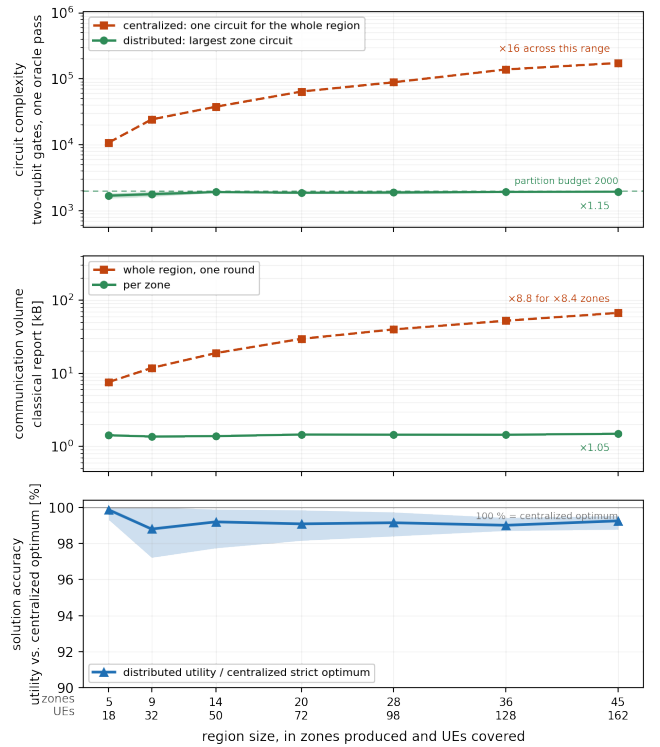


Fig. 10: Region-scale performance.

the hardware requirement follows deployment density rather than region size.

For zone z with n_z^{re} re-samples, coordination traffic is modeled as $(1 + n_z^{\text{re}})K_z(\sum_{i \in B_z} \lceil \log_2 |C_i| \rceil + 32)$ bits, where the 32-bit field carries J_z for the reconstruction of Appendix B-B; the plotted values are converted to kilobytes. Regional traffic

grows by $8.8\times$ as the zone count grows by $8.4\times$, so the mean report per zone increases by only 5% and the per-zone load remains nearly unchanged as the region expands.

Coordinated utility stays between 98.81% and 99.87% of J^* with no downward trend, and all 40 runs finish globally feasible, so (13) holds. Decomposition, finite sampling, and commitment cost about one percentage point, which does not accumulate over the evaluated range.

The exponent backoff keeps sampling affordable. On a separate 16-instance comparison, the most tightly constrained zones need 10^5 to 3×10^7 executions at the target exponent with $k = 1$; capping each zone at 10^4 cuts the peak single-zone requirement from 2.9×10^7 to 10^4 , a factor of about 2900, while mean utility moves from 99.31% to 99.28% of J^* , a loss of 0.03 percentage points on average and 0.55 at worst. Reconstruction is unbiased, so the cap trades effective sample size for execution cost rather than systematic bias.

VI. CONCLUSION

This paper establishes a unified quantum framework for scalable network management through common assignment modeling, direct constraint evaluation, and distributed coordination. Diverse inter-cell and intra-cell management tasks are represented within the same assignment structure while preserving their respective objectives and resource constraints. The quantum computation directly evaluates the original constraints and represents real-valued utilities through utility-dependent measurement probabilities, avoiding penalty reformulation and objective quantization. Zone-based decomposition extends the framework to network-scale problems under limited NISQ resources, while coordination of feasible local quantum outputs constructs a consistent network-wide assignment. Qiskit-based simulations and quantum-processor experiments demonstrate feasible solution generation and the scalability of the proposed computation. These results establish a common quantum computational approach for heterogeneous network assignment problems under present-day quantum resource limitations.

APPENDIX A EXPONENTIAL UTILITY WEIGHT AND ROTATION REALIZATION

A. Uniqueness of the exponential weight

The zone decomposition in (15) expresses the network-wide utility as a sum of local utilities. To preserve this additive structure when the corresponding quantum weights are combined multiplicatively across zones, their product should depend on the local utilities only through their sum.

Let $f : \mathbb{R} \rightarrow (0, \infty)$ be a continuous local weight function for $|\mathcal{Z}| \geq 2$. Suppose that

$$\prod_{z \in \mathcal{Z}} f(J_z) = F\left(\sum_{z \in \mathcal{Z}} J_z\right), \quad (47)$$

where F is strictly increasing. Consider two local utilities J_1 and J_2 , with all other zone utilities set to zero. Their product weight is $f(J_1)f(J_2)f(0)^{|\mathcal{Z}|-2}$. Assigning the same

total utility $J_1 + J_2$ to a single zone gives $f(J_1 + J_2)f(0)^{|\mathcal{Z}|-1}$. Since both cases have the same total utility, (47) gives

$$\frac{f(J_1)f(J_2)}{f(0)} = f(J_1 + J_2). \quad (48)$$

Define the normalized weight as $\tilde{f}(J) = f(J)/f(0)$. Then, (48) is expressed as

$$\tilde{f}(J_1)\tilde{f}(J_2) = \tilde{f}(J_1 + J_2). \quad (49)$$

The continuous positive solutions of this exponential Cauchy equation have the form $\tilde{f}(J) = e^{\lambda J}$ for a real constant λ , and the corresponding local weight is

$$f(J) = f(0)e^{\lambda J}, \quad (50)$$

and the strict increase of F gives $\lambda > 0$. The exponential form is the unique continuous positive multiplicative weight, up to an assignment-independent factor, that preserves the additive utility relation under the zone decomposition.

B. Realization by controlled rotations

The exponential weight is realized through the utility-qubit rotations defined in (21). For a complete local assignment $\mathbf{z}_z$, the assignment qubits specify the selected candidates, and the corresponding controlled rotations determine the probabilities of the utility qubits. According to (22), selecting candidate j for UE i gives probability $g_{i,j}$ of observing its utility qubit in $|0\rangle$. The probability that all utility qubits satisfy this condition gives the zone-level utility weight $g_z(\mathbf{z}_z)$ defined in (38). Using the exponential mapping in (20), the zone-level weight can be expressed as

$$g_z(\mathbf{z}_z) = \kappa_z \exp[\lambda J_z(\mathbf{z}_z)], \quad (51)$$

where $\kappa_z > 0$ is independent of the selected local assignment. The assignment-independent factor disappears when the probabilities over feasible assignments are normalized. The resulting conditional distribution becomes (40). Also, zone-level weights preserve the network-wide utility under the decomposition. From (15),

$$\prod_{z \in \mathcal{Z}} g_z(\mathbf{z}_z) \propto \exp\left[\lambda \sum_{z \in \mathcal{Z}} J_z(\mathbf{z}_z)\right] = \exp[\lambda J(\mathbf{z})]. \quad (52)$$

The controlled rotations realize the multiplicative exponential weighting established above while preserving the original network-wide utility represented by the zone decomposition.

APPENDIX B COORDINATION OF ZONE-LOCAL ASSIGNMENTS

A. Reduction to boundary assignments

Let $\mathcal{B}_z \subseteq \mathcal{U}_z$ denote the boundary UEs represented in zone z . For a local assignment $\mathbf{z}_z$, $\mathbf{z}_{\mathcal{B}_z}$ denotes its restriction to these boundary UEs, while the assignments of the remaining UEs are internal to the zone. Network-wide coordination concerns only the boundary assignments since the internal assignments do not appear in the consistency conditions between zones. From (13), (15), and (16), the zone-local distributions in

(40) recover the network-wide distribution when the boundary assignments are consistent across zones as

$$p(\mathbf{z}) \propto \prod_{z \in \mathcal{Z}} p_z(\mathbf{z}_z) \prod_{i \in \mathcal{B}} \delta_i(\mathbf{z}_i), \quad (53)$$

where $\delta_i(\mathbf{z}_i)$ is the consistency indicator in (12). For a boundary assignment $\mathbf{s}$, the probability mass of the feasible local assignments consistent with $\mathbf{s}$ is

$$p_z^b(\mathbf{s}) = \sum_{\mathbf{z}_z \in \{\mathbf{x} \in \mathcal{F}_z : \mathbf{x}_{\mathcal{B}_z} = \mathbf{s}\}} p_z(\mathbf{z}_z). \quad (54)$$

Marginalizing (53) over the internal assignments gives

$$p_{\mathcal{B}}(\mathbf{z}_{\mathcal{B}}) \propto \prod_{z \in \mathcal{Z}} p_z^b(\mathbf{z}_{\mathcal{B}_z}). \quad (55)$$

Proposition 2 (Boundary reduction). *The reduction in (54) preserves the network-wide distribution over the boundary assignments. Once the boundary assignments are fixed consistently across zones, the internal assignments can be determined independently within their respective zones.*

Proof: For a fixed consistent boundary assignment, all consistency indicators δ_i equal one. The remaining factors in (53) depend only on their respective zone-local variables. Marginalizing each factor over its internal assignments gives (54), and their product gives (55). ■

Proposition 2 reduces network-wide coordination to the boundary assignments. Each zone estimates the joint boundary distribution in (54) from its accepted samples as

$$\hat{p}_z^b(\mathbf{s}) = \frac{|\{\mathbf{z}_z \in \mathcal{S}_z : \mathbf{z}_{\mathcal{B}_z} = \mathbf{s}\}|}{|\mathcal{S}_z|}. \quad (56)$$

The resulting distribution preserves the relationship among boundary assignments represented by the joint local samples. The UE-wise marginals used for confidence-ordered coordination are obtained from these same samples as in (44).

B. Reweighting of zone-local samples

The sampling exponent can differ across zones according to their local quantum configurations. Let $\lambda_z \leq \lambda$ denote the exponent used for sampling in zone z , where λ is the common exponent used for coordination. Let $\mathbf{z}_z^{(n)}$ denote the n -th accepted sample in that zone. Under local sampling, its relative probability is proportional to $\exp[\lambda_z J_z(\mathbf{z}_z^{(n)})]$, whereas the common exponent assigns a relative weight proportional to $\exp[\lambda J_z(\mathbf{z}_z^{(n)})]$. Their ratio gives the reweighting factor as

$$w_z^{(n)} = \frac{\exp[\lambda J_z(\mathbf{z}_z^{(n)})]}{\exp[\lambda_z J_z(\mathbf{z}_z^{(n)})]} = \exp[(\lambda - \lambda_z) J_z(\mathbf{z}_z^{(n)})]. \quad (57)$$

Feasibility is evaluated independently of sampling exponent, and the feasible set $\mathcal{F}_z$ is unchanged. The weights in (57) adjust the empirical local distribution from the sampling exponent λ_z to the common coordination exponent λ . A larger difference between λ and λ_z can concentrate the reweighted probability mass on a smaller number of accepted samples. This concentration is measured by the effective sample size

$$\tilde{K}_z = \frac{(\sum_{n=1}^{K_z} w_z^{(n)})^2}{\sum_{n=1}^{K_z} (w_z^{(n)})^2}, \quad (58)$$

where $K_z = |\mathcal{S}_z|$. The effective sample size equals K_z when the sample weights are uniform and approaches one when a single sample dominates the total weight. Since $\tilde{K}_z \leq K_z$, $\tilde{K}_z$ provides the sample sufficiency measure used during coordination. The reweighting factors in (57) give the sample weights in the empirical marginal (44). The joint boundary distribution $\hat{p}_z^b$ in (56) is reweighted in the same manner by summing the weights of the samples associated with each boundary assignment. When $\lambda_z = \lambda$, all $w_z^{(n)} = 1$, and weighted estimates reduce to their unweighted empirical forms.

C. Confidence-ordered decimation

Confidence-ordered decimation resolves the boundary assignments one UE at a time while preserving the assignment relationships contained in the joint local samples. At the t -th iteration, let $\mathcal{S}_z^{(t)}$ denote the current sample set in zone z , and let $\pi_{z,i}^{(t)}(j)$ denote the corresponding marginal probability for an unresolved boundary UE i . The marginals from zones containing UE i obtain the consensus distribution as

$$b_i^{(t)}(j) = \frac{\prod_{z \in \mathcal{Z}_i} \pi_{z,i}^{(t)}(j)}{\sum_{j' \in \mathcal{C}_i} \prod_{z \in \mathcal{Z}_i} \pi_{z,i}^{(t)}(j')}. \quad (59)$$

The concentration of this distribution is measured using the normalized entropy defined as

$$\gamma_i^{(t)} = 1 + \frac{\sum_{j \in \mathcal{C}_i} b_i^{(t)}(j) \log b_i^{(t)}(j)}{\log |\mathcal{C}_i|}. \quad (60)$$

A larger $\gamma_i^{(t)}$ indicates a stronger preference for a particular candidate. Note that a uniform consensus gives $\gamma_i^{(t)} = 0$. Among the unresolved boundary UEs, the UE with the largest confidence is selected as

$$i^* = \arg \max_{i \in \mathcal{B}_u^{(t)}} \gamma_i^{(t)}, \quad (61)$$

where $\mathcal{B}_u^{(t)}$ denotes the set of unresolved boundary UEs at the t -th iteration. The candidate with the largest consensus probability for the selected UE is

$$j^* = \arg \max_{j \in \mathcal{C}_{i^*}} b_{i^*}^{(t)}(j). \quad (62)$$

UE i^* is fixed to candidate j^* . For each zone containing this UE, the joint sample set is filtered according to

$$\mathcal{S}_z^{(t+1)} = \left\{ \mathbf{z}_z \in \mathcal{S}_z^{(t)} : z_{i^*,j^*} = 1 \right\}. \quad (63)$$

The sample sets of the other zones remain unchanged.

Since each sample specifies a complete local assignment, filtering one boundary UE also changes empirical candidate distributions of unresolved boundary UEs in the affected zones. Their marginal, consensus, and confidence values are recomputed from the filtered sample sets before the next boundary UE is selected. Applying the filtering operation to the joint samples, rather than directly modifying their marginals, preserves the assignment relationships represented by local quantum outputs. Filtering can reduce the effective sample size $\tilde{K}_z$. If $\tilde{K}_z < K_{\min}$ in an affected zone, that zone is sampled again with its already resolved boundary assignments fixed during state preparation. Only the affected zone is

Algorithm 1 Confidence-ordered decimation

```

1:  $t \leftarrow 0$ 
2:  $\mathcal{S}_z^{(0)} \leftarrow \mathcal{S}_z$  for all  $z \in \mathcal{Z}$ 
3:  $\mathcal{B}_u^{(0)} \leftarrow \mathcal{B}$ 
4: while  $\mathcal{B}_u^{(t)} \neq \emptyset$  do
5:   compute  $\pi_{z,i}^{(t)}$  from the current weighted samples
6:   compute  $b_{z,i}^{(t)}$  by (59)
7:   compute  $\gamma_i^{(t)}$  by (60)
8:   select  $i^*$  by (61)
9:   select  $j^*$  by (62)
10:  fix UE  $i^*$  to candidate  $j^*$ 
11:  filter the affected sample sets by (63)
12:  if  $\bar{K}_z < K_{\min}$  for an affected zone  $z$  then
13:    re-sample zone  $z$  with its resolved boundary assignments fixed
14:  end if
15:   $\mathcal{B}_u^{(t+1)} \leftarrow \mathcal{B}_u^{(t)} \setminus \{i^*\}$ 
16:   $t \leftarrow t + 1$ 
17: end while

```

sampled again, and the sample sets of the other zones remain unchanged. In practice, a small pseudo-count can be added when estimating empirical marginals to prevent a candidate that is absent from a finite sample set from producing a zero consensus probability. The procedure terminates after at most $|\mathcal{B}|$ decimation steps, when all boundary UEs have consistent assignments across their corresponding zones. The remaining local samples satisfy their zone-level feasibility conditions, and the resolved boundary assignments satisfy the consistency conditions in (13). Algorithm 1 summarizes the overall procedure.

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